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AHB-Lite Master, Slave and Wrapper Design

A Verilog Implementation of the AMBA 3 AHB-Lite Protocol

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1. Introduction

This report documents a Verilog implementation of a single-master, single-slave system based on the AMBA 3 AHB-Lite protocol, as defined in the ARM specification ARM IHI 0033A. The design consists of three modules: an AHB-Lite master (ahb_lite_master), an AHB-Lite slave (ahb_lite_slave), and a top-level wrapper (ahb_system_top) that connects the master and slave together and exposes a simple command interface and a testbench control signal.

The purpose of this report is to describe the design, explain how it maps onto the AHB-Lite specification, and clearly state which protocol features the design correctly implements and which features it does not handle.

2. AHB-Lite Protocol Overview

AHB-Lite is a single-master bus interface intended for high-performance, high clock frequency designs. A transfer consists of an address phase (one clock cycle, unless extended by the previous transfer) and a data phase (one or more cycles, extended using HREADY). The master drives address and control signals; the slave drives read data, HREADYOUT and HRESP. Figure 1 below, taken from the AHB-Lite specification, shows the general block structure of an AHB-Lite system with a master, an address decoder, a set of slaves, and a read-data multiplexor.

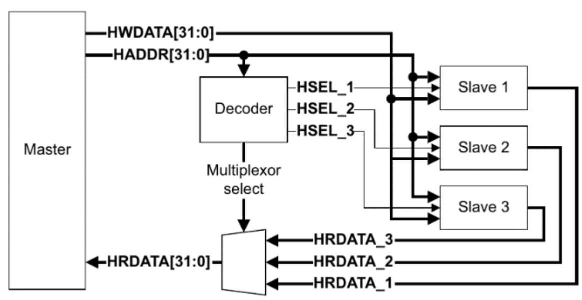


Figure 1. AHB-Lite system block diagram (Source: ARM IHI 0033A, Figure 1-1)

Every transfer is classified by HTRANS as IDLE, BUSY, NONSEQ or SEQ, and every burst is classified by HBURST as SINGLE, an undefined-length INCR burst, or one of the fixed-length INCR/WRAP bursts of 4, 8 or 16 beats. Figure 2 shows the simplest case: a single read and a single write transfer with no wait states.

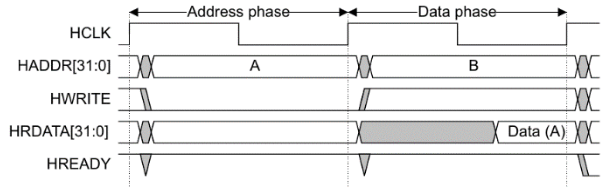


Figure 3-1 Read transfer

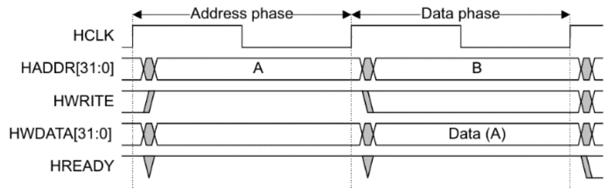


Figure 3-2 Write transfer

Figure 2. Basic read and write transfers (Source: ARM IHI 0033A, Figures 3-1 and 3-2)

3. Design Description

3.1 ahb_lite_master

The master is driven by a simple user command interface (cmd_valid, cmd_addr, cmd_write, cmd_size, cmd_burst, cmd_lock, cmd_wdata, cmd_busy, cmd_ready) rather than directly by user logic driving AHB signals. Internally it is built around a four-state FSM (IDLE, BUSY, NONSEQ, SEQ) whose state encoding is deliberately made identical to the HTRANS[1:0] encoding defined by the protocol, so HTRANS is generated almost directly from the state register. A combinational 'zero-latency bypass' drives HADDR/HWRITE/HSIZE/HBURST/HMASTLOCK from the incoming command on the first (NONSEQ) cycle of a transfer, and from internal registers for all following beats. A dedicated function, calc_next_addr, reproduces the address-wrapping arithmetic described in the specification for WRAP4/WRAP8/WRAP16 bursts, and simple incrementing arithmetic for INCR-type bursts.

3.2 ahb_lite_slave

The slave implements a small synchronous memory (default 256 words). It samples HADDR, HWRITE and transfer validity on every cycle in which HREADY is high, performs the memory write in the following cycle when the transfer is valid, and drives HRDATA from the sampled address. HREADYOUT is driven from an external force_wait test input (used to inject wait states from a testbench) rather than from any internal wait-state condition, and HRESP is tied permanently to OKAY (0).

3.3 ahb_system_top (wrapper)

The wrapper instantiates one master and one slave and connects them directly: the slave's HREADYOUT drives the shared HREADY net, which feeds back into both the master and the slave, and HRDATA/HRESP pass straight from the slave to the master. Because the system contains only one slave, no address decoder and no read-data multiplexor are instantiated; the wrapper exposes the command interface and the slv_force_wait test signal at the top level.

4. Features the Design Can Handle

Based on comparison against the AHB-Lite specification, the design correctly implements the following:

Feature	Notes
Basic single read and write transfers	Address phase / data phase pipelining, matching Figures 3-1 and 3-2 of the specification.
Wait-state insertion via HREADY	Both master and slave correctly freeze state while HREADY is low.
SINGLE burst	HBURST = 3'b000.
Fixed-length bursts WRAP4 / INCR4	HBURST = 3'b010 / 3'b011, correct address-wrap arithmetic.
Fixed-length bursts WRAP8 / INCR8	HBURST = 3'b100 / 3'b101.
Fixed-length bursts WRAP16 / INCR16	HBURST = 3'b110 / 3'b111.
HTRANS encoding (IDLE / BUSY / NONSEQ / SEQ)	Generated directly from the FSM state, matching Table 3-1 of the specification.
Mid-burst BUSY insertion	cmd_busy allows the master to insert BUSY transfers between beats of a burst, holding address/control for the next beat as required.
HMASTLOCK signalling	Passed through from cmd_lock to the bus.
Two-cycle ERROR handling on the master side	The master correctly stays in place during the first ERROR cycle (HREADY low) and returns to IDLE once HREADY goes high, aborting the remaining burst.
Basic clock/reset behaviour	Single clock (HCLK), asynchronous active-low reset (HRESETn), consistent with Chapter 7 of the specification.

5. Features the Design Cannot Handle

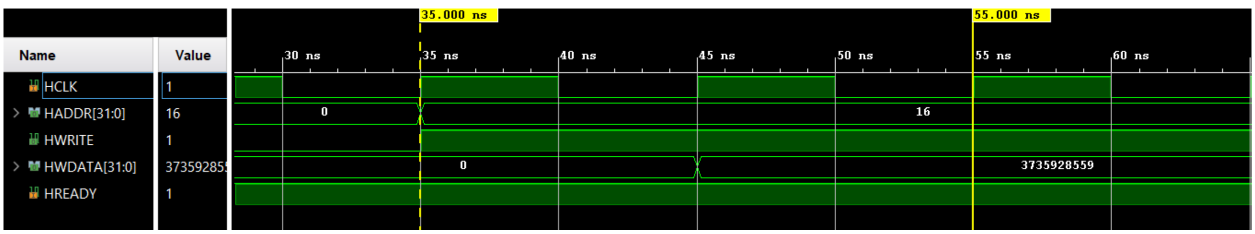
The following protocol features, present in the AHB-Lite specification, are either not implemented or not fully compliant in this design:

Feature / Limitation	Explanation
Undefined-length INCR burst (HBURST = 3'b001)	Not decoded by the master's FSM; falls into the default case and is treated as a single, one-beat transfer instead of an open-ended burst.

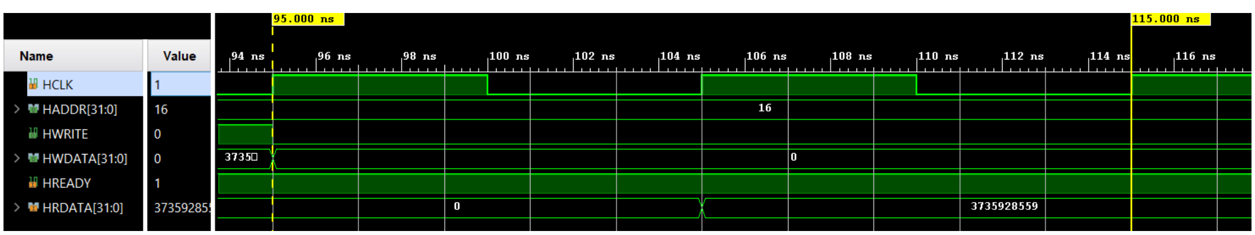
HPROT protection signals	No HPROT port exists on the master or slave; opcode/data, privileged/user and cacheable/bufferable information is not generated or used.
Address decoding and multi-slave interconnect	No address decoder (HSELx generation) and no read-data/response multiplexor are present. Only one hard-wired slave is supported; the multi-slave structure shown below cannot be built with this design as-is.
Byte- and halfword-size transfers	HSIZE is accepted as an input but is not used to build byte-lane write enables. A byte or halfword write will overwrite the entire 32-bit word rather than only the addressed bytes, and no endianness scheme is implemented.
Data bus widths other than 32 bits	Only a 32-bit HWDATA/HRDATA path is implemented; the specification's wider options (64 through 1024 bits) are not supported.

6. Simulation Results

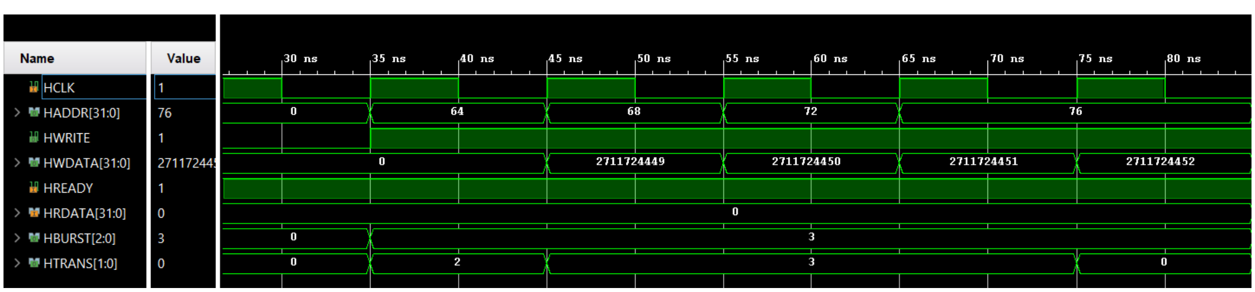
6.1. write operation



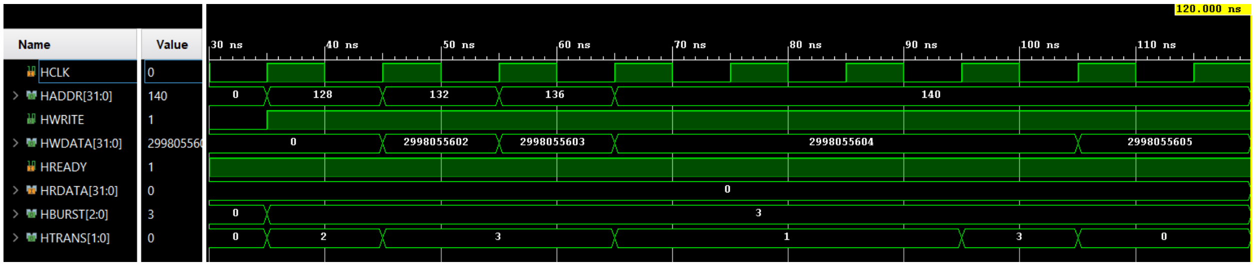
6.2. read operation (after first write)



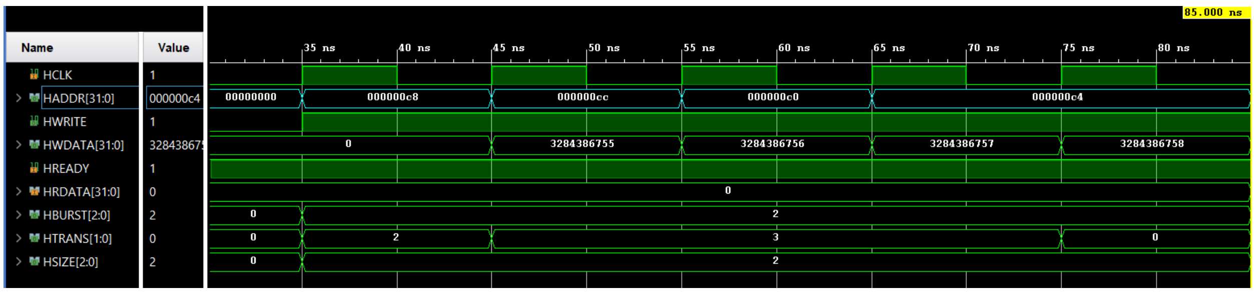
6.3. write INCR4



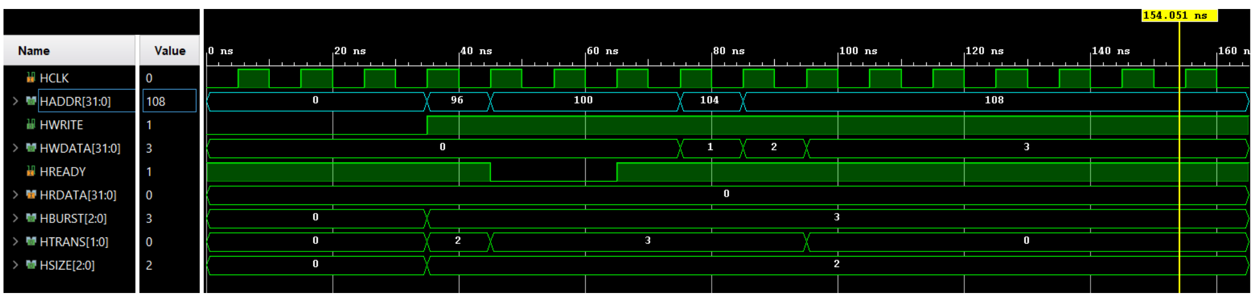
6.3. write INCR4 with busy



6.4.write WRAP4



6.5. slave is busy



7. Conclusion

The design implements the core pipelined transfer mechanism of AHB-Lite, together with all fixed-length burst types, mid-burst BUSY insertion and basic wait-state and error handling, for a single master and a single 32-bit-wide slave. It does not implement protection signalling (HPROT), byte/halfword write strobes, multi-slave address decoding and read-data multiplexing, data bus widths above 32 bits, or genuinely slave-generated wait states and error responses. These limitations should be kept in mind before reusing the design outside of a simple, single-slave, word-aligned test environment.